

How to use simulator

input

- IR_MAP.csv → Ri,Rp,C of target cell (see part-1 parameter identification)
- bat_pack_DOE_initial_condition.csv → Initial conditions of battery pack (you can change with GUI)
- drive_condition.csv → Time-series output profile

src

- Batt_sim_main.py → DOE engine (see part-2 Battery-energy-simulation)
- DOE_bat_pack.py → just run this file
- bat_pack_cond_loader.py
- bat_sequential_calc.py
- cell_const_calc.py
- interpolate_cell.py
- ir_map_load.py
- load_drive_condition.py

Place all of these files into a single folder. Please update the `csv\_path` in `DOE\_bat\_pack.py` to point to that folder's location.

Output file

output

bat_pack_DOE_applied_conditions.csv

bat_sim_DOE_result.csv

bat_sim_conclusion.csv

bat_sim_result.csv

Residual artifacts of the final DOE iteration

Battery pack condition

cell capa(Ah)	1.8	
cell wt g	100.0	
	max	min
n-seri	200	100
n-para	4	4
start soc	50.0	

threshold

system max current A	300.0
cell max charge current A	300.0
system max voltage V	700.0
system min voltage V	290.0
cell max lmt V	4.2
cell min lmt V	2.7

start simulation

1	cell capa(Ah)	1.8
2	wt g	100.0
3	max v	4.2
4	min v	2.7
5	max charge cell cur	300.0
6	start soc	50.0
7	max current	300.0
8	max sys v	700.0
9	min sys v	290.0
10	n-seri max	200.0
11	n-seri min	100.0
12	n-para max	4.0
13	n-para min	1.0

1	n-seri	n-para	result	finished soc	pack weight kg	DOD	max cell voltage	min cell voltage	total loss wh
2	190	4	Passed	49.78574169658846	76.0	2.7479720851318845	3.6514782487019692	3.4699832679439897	29.162675267986554
3	188	4	Passed	49.78213603610847	75.2	2.7779095030428635	3.6521563045223955	3.468691122755914	29.483847853181974
4	186	4	Passed	49.77842318061912	74.4	2.8085069839370576	3.65284871075256	3.4673700260203235	29.812196509297888
5	184	4	Passed	49.774598626198	73.6	2.839786621244663	3.653555927956982	3.466018992858807	30.147965238276615
6	182	4	Passed	49.770657610100955	72.8	2.8717715064960245	3.654278436627224	3.4646369931683814	30.491418307034643
7	180	4	Passed	49.76659513551607	72.0	2.904485786386843	3.65501673827169	3.463222948993008	30.84281525202958
8	178	4	Passed	49.76240591653702	71.2	2.9379547238082537	3.6557713565777576	3.46177573170946	31.20243501426924
9	176	4	Passed	49.75808437025574	70.4	2.9722047631666726	3.6565428386518892	3.4602941590119958	31.57057070940808
10	174	4	Passed	49.75362459959876	69.6	3.0072636003505266	3.6573317563439436	3.4587769916789126	31.947529707634057
11	172	4	Passed	49.7490203567423	68.8	3.043160257734364	3.6581387076623755	3.4572229301022315	32.33363404808287
12	170	4	Passed	49.74426503583743	68.0	3.079925164649275	3.658964318287745	3.4556306105600796	32.729221897152094
13	168	4	Passed	49.739351641026246	67.2	3.1175902437897065	3.65980924319256	3.4539986012091632	33.134648364433566